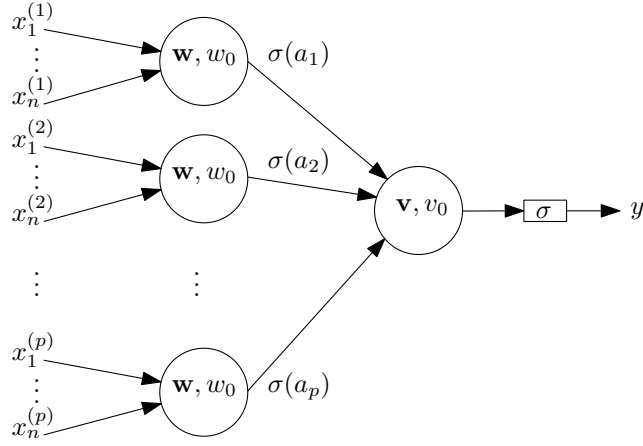


COMPUTER SCIENCE TRIPOS Part IB – 2022 – Paper 6

2 Artificial Intelligence (sbh11)

This question addresses a variation on the usual *multilayer perceptron*.



The input vector is divided into p groups each with n elements. Let $x_i^{(j)}$ be the i th element in the j th group and let $\mathbf{x}^{(j)} = (x_1^{(j)} \dots x_n^{(j)})^T$. There are p nodes in the hidden layer, which share a single weight vector $\mathbf{w}$ and bias w_0 . Thus the k th hidden node computes $\sigma(a_k)$ where σ is an activation function and

$$a_k = \mathbf{w}^T \mathbf{x}^{(k)} + w_0.$$

Let $\mathbf{a} = (a_1 \dots a_p)^T$. The output node then combines the hidden nodes in the usual way using weights $\mathbf{v}$ and v_0 to produce $y = \sigma(a)$ where

$$a = \sum_{i=1}^p v_i \sigma(a_i) + v_0.$$

Collecting all the parameters of the network into a single vector $\boldsymbol{\theta}$, the error for a single labelled example is $E(\boldsymbol{\theta})$.

Backpropagation
for the MLP

(a) Show that the value of $\delta = \partial E(\boldsymbol{\theta}) / \partial a$ is

$$\delta = \sigma'(a) \frac{\partial E(\boldsymbol{\theta})}{\partial y}.$$

[3 marks]

Answer: This is unchanged from the usual MLP derivation. As

$$\theta = \frac{\partial E(\boldsymbol{\theta})}{\partial y} \frac{\partial y}{\partial a}$$

and $y = \sigma(a)$ the result follows directly.

Backpropagation
for the MLP

(b) Find expressions for the partial derivatives of $E(\boldsymbol{\theta})$ with respect to the

parameters of the single output node.

[5 marks]

Answer: This is unchanged from the usual MLP derivation. As

$$\frac{\partial E(\boldsymbol{\theta})}{\partial v_i} = \frac{\partial E(\boldsymbol{\theta})}{\partial a} \frac{\partial a}{\partial v_i}$$

we have

$$\frac{\partial E(\boldsymbol{\theta})}{\partial v_i} = \delta \sigma(a_i)$$

for $i = 1, \dots, p$ and

$$\frac{\partial E(\boldsymbol{\theta})}{\partial v_0} = \delta.$$

Backpropagation
for the MLP

(c) Show that the partial derivatives $\delta_i = \partial E(\boldsymbol{\theta}) / \partial a_i$ for the hidden nodes are

$$\delta_i = \delta v_i \sigma'(a_i).$$

[5 marks]

Answer: This is unchanged from the usual MLP derivation. As

$$\delta_i = \frac{\partial E(\boldsymbol{\theta})}{\partial a} \frac{\partial a}{\partial a_i} = \delta \frac{\partial a}{\partial a_i}$$

and $a = \sum_j v_j \sigma(a_j) + v_0$ we have

$$\frac{\partial a}{\partial a_i} = v_i \sigma'(a_i)$$

and the result follows.

Backpropagation
in a more general
case

(d) Find expressions for the partial derivatives of $E(\boldsymbol{\theta})$ with respect to the parameters of the hidden nodes. [7 marks]

Answer: This requires a small change to the usual MLP derivation because the weights are shared among the nodes. Changing any of the weights $\mathbf{w}, w_0$ changes all the activations in $\mathbf{a}$. So

$$\frac{\partial E(\boldsymbol{\theta})}{\partial w_i} = \sum_{j=1}^p \frac{\partial a_j}{\partial w_i} \frac{\partial E(\boldsymbol{\theta})}{\partial a_j}.$$

The second of these was derived in the previous part, and as

$$a_j = \mathbf{w}^T \mathbf{x}^{(j)} + w_0$$

we have

$$\frac{\partial a_j}{\partial w_i} = x_i^{(j)}$$

when $i = 1, \dots, p$ or 1 when $i = 0$. Thus

$$\frac{\partial E(\boldsymbol{\theta})}{\partial w_i} = \sum_{j=1}^p \delta_j x_i^{(j)}$$

when $i = 1, \dots, p$ or $\sum_{j=1}^p \delta_j$ when $i = 0$.
